

Trying to read and write SD cards using 6809 assembly language

 flexonsbd.blogspot.com/2022/06/sd6809.html

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Trying to read and write to an SD card using 6809 assembly language.

I tried reading and writing to an SD card in SPI mode using the 6809 assembler.

I've been interested in SD cards for a while, but they seemed complicated, so I never got around to trying them. Last year, I tried reading and writing to sensors in SPI mode, and I was impressed by how much easier it was than I2C.

"Controlling sensors via SPI communication using assembler from a 6821PIA (November 3, 2021)"

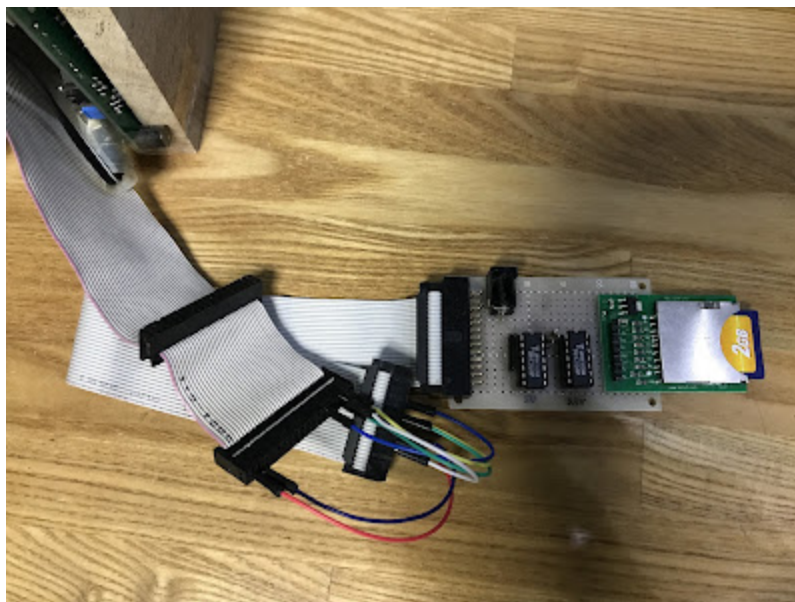
Based on that experience, I decided to try reading and writing to an SD card in SPI mode, thinking it might be relatively easy.

I referenced the blog of Mr. Hishida from Chunichi Denko, who I always rely on, titled " [\[New Series\] The Great Microcontroller Independence Project: Creating an SD Card Interface](#) " Parts 1-32. Thank you as always for the valuable information.

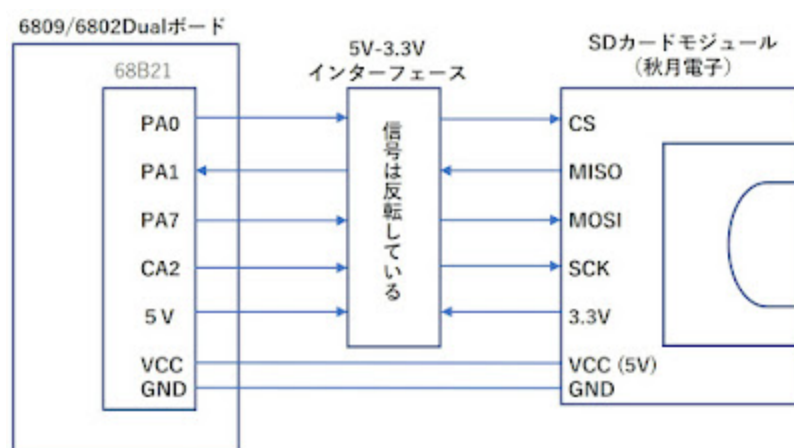
So, I used the same 5V-3.3V conversion circuit from the 8th blog post, and while the original article uses an 82C55 for control, I used the 68B21 found in the 6809/6802 Dual microcontroller.

The image and connection diagram of the experimental circuit are shown.

The board standing upright on the left of the image is a 6809/6802 Dual board, connected to the experimental circuit via a flat cable from the 68B21.



Experimental circuit



Block diagram

Port A of the 68B21 has a convenient feature that allows it to function as both an output and an input, so I'm using only Port A for input and output. The clock SCK uses the 1-bit output CA2. In my program, the clock frequency from CA2 was approximately 30.7 kHz.

Since the 5V-3.3V interface is composed of a single inverter, the signal will be inverted, so this will need to be taken into consideration in the program.

Akizuki Denshi's SD card module is designed to generate 3.3V from a 5V power supply, so that 3.3V is also supplied to the interface circuit.

Program creation

I created the program on Windows.

I create the source code using a Windows editor (I use EmEditor), assemble it using a custom 6809 cross-assembler that runs on Windows, and then load and execute the resulting mot file using the L command on the 6809/6802 Dual board monitor.

I will show you the steps with images.

Create the program listing in the editor.



```
1 #
2 * SD-RW.TXT
3 #
4 * SD Read/Write on SBC
5 #
6 * 6821(P1A) Port A <--> SD
7 #
8 * SBC SD
9 * CA2 2 --> 16 SD
10 * PA0 3 --> 15 CS
11 * PA1 4 <-- 7 MOSI
12 * PA7 10 --> 20 MISO
13 * QND 31 -- 10 QND
14 * VCC 33 -- 9 VCC
15 #
16 * cf. all data line is inverted
17 #
18 * P1 output mode
19 * CA2 1bit output
20 #
21 * READ/WRITE BUFFER: $0000 - $7FFF
22 #
23 * Original version "2022.06.11"
24 * Copyright by Y.Yamamoto
25 #
26 MON EQU $F800
27 INCH EQU $F000
28 OUTCH EQU $F002
29 PSTING EQU $FAM
30 #
31 RWBUF EQU $0000
32 #
33 PADAT EQU $F00C
34 PACTL EQU PADAT+1
35 #
36 ORG $0000
37 #
38 START BRA MAIN
39 NV FCB 1
40 SCTN0 FCB 0 top adrs of sct
41 SCTN1 FCB 0
42 SCTN2 FCB 0
43 SCTN3 FCB 0
44 RESDAT FCB 0 response data
```

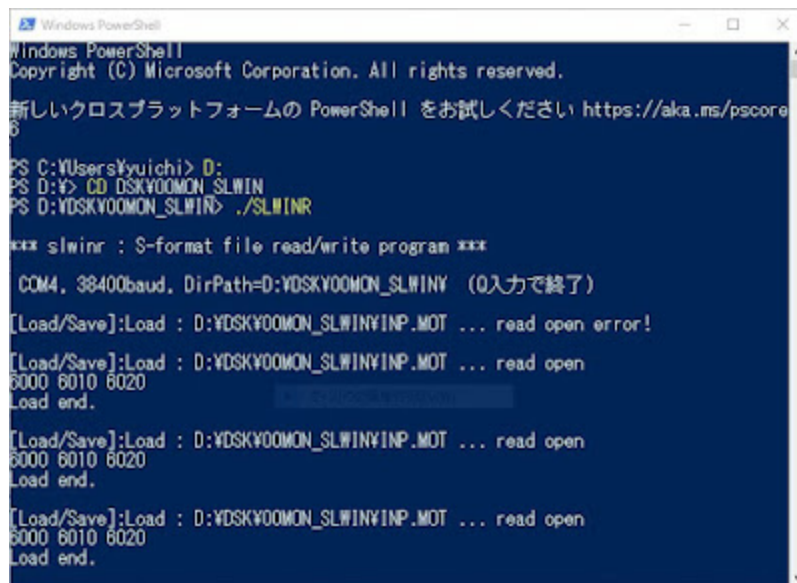
Creating in the editor

The completed program is then fed into an assembler and assembled.



Assembling

Load the program using the L command on the execution screen. Normally, you only need to run this loading program once and do not need to interact with it further, so you can close it.



Loading program in progress

This is the execution screen. Load the file with the L command and execute it with the G command.

Loading and running

If the program needs modification, repeat the above steps.

I encountered quite a few problems before I was actually able to read and write to the SD card, and in addition to the blog of Chunichi Denko, I also referred to the blogs of "[Elm by ChaN](#)," "[SD Card](#)," and "[SD Card Operation](#)." Thank you very much.

The program is currently complete, and it comes in two versions: one that can read and write to both SD and SDHC cards, and another that can only read and write to SD cards.

Since the goal is to integrate it into the Flex9 6809/6802 Dual board, it's best to make it as short as possible, so I plan to make it SD-only. However, since I had the opportunity, I also checked to make sure it could read and write SDHC cards as well.

This section lists programs compatible with both SD and SDHC cards. The only difference from the SD-only version is the method of specifying sector numbers in the initialization routine and the Read/Write routine; the initialization routine, in particular, is considerably more complex than the SD-only version.

```

0001 *          0002 * SDHC - RW . TXT          0003 *
0004 * SD Read / Write on SBC          0005 *          0006 * 6821 ( PIA
) Port A <--> SD          0007 *          0008 * SBC SD
0009 * CA2 2 --> 16 SCK          0010 * PA0 3 --> 15 CS
0011 * PA1 4 <-- 7 MISO          0012 * PA7 10 --> 20 MOSI
0013 * GND 31 -- 10 GND          0014 * VCC 33 -- 9 VCC
0015 *          0016 * cf . all data line is inverted          0017 *
0018 * PA output mode          0019 * CA2 1-bit output
0020 *          0021 * READ / WRITE BUFFER $6600 - 67FF          0022
*          0023 * Original version '2022.06.11          0024 *
Copyright by Y.Yamamoto0025 *0026 F800 MON EQU $F800          0027 F000 INCHA EQU

```

```

$F000          0028 F002 OUTCHA EQU $F002          0029 FAA4 PSTRNG EQU $FAA4
0030          0031 6600 RWBUF EQU $6600          0032          0033 F0DC
PADAT EQU $F0DC          0034 F0DD PACTL EQU PADAT+1          0035          0036
6000 ORG $6000          0037          0038 6000 20 07 START BRA MAIN
0039 6002 01 VN FCB 1          0040 6003 00 SCTN0 FCB 0 top adrs of sct0041 6004
00 SCTN1 FCB 0          0042 6005 00 SCTN2 FCB 0          0043 6006 00 SCTN3
FCB 0          0044 6007 00 RESDAT FCB 0 response data0045 6008 00 SDHC FCB 0
sd:0,sdhc:10046          0047 6009 MAIN EQU *          0048 *=====reset
pia=====0049 *portA:output0050 6009 86 30 LDA #$30          0051 600B B7 F0DD STA
PACTL select ddra, ca2:00052 600E 86 FF LDA #$FF          0053 6010 B7 F0DC STA
PADAT set output mode0054 6013 86 34 LDA #$34          0055 6015 B7 F0DD STA PACTL
select peripheral,ca2:00056          0057 *for pia exist check0058 6018 86 55
LDA #$55 test data0059 601A B7 F0DC STA PADAT          0060 601D B6 F0DC LDA PADAT
0061 6020 81 55 CMPA #$55          0062 6022 27 09 BEQ INIT          0063 6024 8E
64A0 LDX #MERR          0064 6027 BD FAA4 JSR PSTRNG          0065 602A 7E F800 JMP
MON          0066          0067 602D INIT EQU *          0068 602D 8D 3A
BSR SDINIT          0069          0070 602F MENU EQU *          0071 602F
BD 6384 JSR CRLF          0072 6032 8E 643F LDX #MSMENU          0073 6035 BD FAA4
JSR PSTRNG          0074          0075 6038 BD F000 JSR INCHA          0076
603B 81 31 CMPA #' 1          0077 603D 26 05          BNE MEN2
0078 603F 17 0199          LBSR SDRD          0079 6042 20          EB
BRA MENU          0080 6044 81 32          MEN2 CMPA          #'2
0081 6046 26 05          BNE MEN3          0082 6048 17 0208
LBSR SDWRT          0083 604B 20          E2 BRA MENU          0084
604D 81 33          MEN3 CMPA          #'3          0085 604F 26 05
BNE MEN4          0086 6051 17 0149          LBSR GETSD
0087 6054 20          D9 BRA MENU          0088 6056 81 34          MEN4 CMPA          #'4
0089 6058 26 05          BNE MEN5          0090 605A 17 000C
LBSR SDINIT          0091 605D 20          D0 BRA MENU          0092
605F 81 39          MEN5 CMPA          #'9          0093 6061 26          CC BNE MENU
0094 6063          BD          6384          JSR CRLF          0095 6066 7E          F800
JMP MON          0096 *          0097          0098 6069
SDINIT EQU          *          0099 *----- start -----
0100 * cs : 1 , di : 1 , sck > 74ms          0101 6069 86          FF LDA          #$FF
mosi:b7=H,miso:b1=H(for PIA),cs:b0=H          0102 606B 43          COMA
0103 606C          B7 F0DC STA PADAT          0104 606F          C6          50
LDB          #80 80times          0105 6071 17 02D5          ISTT LBSR SETCKL
0106 6074 17 02DB          LBSR SETCKH          0107 6077 5A
DECB          0108 6078 26          F7 BNE ISTT          0109 *
0110 *----- cmd 0 -----          0111 607A          SCMD0
0112 607A 17 02CC          LBSR SETCKL          0113 607D          C6
06          LDB          #6          0114 607F 8E 6409          LDX
#CMD0          0115 6082          A6          80          SC01 LDA          , X +
0116 6084 43          COMA          0117 6085 17 0250
LBSR SNDDAT          0118 6088 5A          DECB
0119 6089 26          F7 BNE SC01          0120 * get response
0121 608B 5F          CLRB          0122 608C          GR00
0123 608C 5A          DECB          0124 608D 1027 0141
LBEQ SDERR          0125 6091 17 0260          LBSR GETDAT

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0126 6094      B6      6007      LDA RESDAT      0127 6097 43
COMA      0128 6098 17 02C0      LBSR DSPBIN disp response
0129 609B 81 01      CMPA      #$01      0130 609D 26      ED
BNE GR00      0131      0132 609F      BD      6384
JSR CRLF      0133 *      0134 *----- cmd 8 -----
0135 60A2      SCMD8      0136 60A2 17 02A4      LBSR
SETCKL      0137 60A5      C6      06      LDB      #6
0138 60A7 8E 6421      LDX      #CMD8      0139 60AA      A6
80      SC80 LDA      , X +      0140 60AC 43      COMA
0141 60AD 17 0228      LBSR SNDDAT      0142 60B0 5A
DECB      0143 60B1 26      F7 BNE SC80      0144 * get
response      0145 60B3      GR80      0146 60B3 17
023E      LBSR GETDAT      0147 60B6      B6      6007
LDA RESDAT      0148 60B9 43      COMA
0149 60BA 17 029E      LBSR DSPBIN disp response      0150 60BD 81 01
CMPA      #$01      0151 60BF 26      F2 BNE GR80
0152 60C1      C6      04      GR81 LDB      #4      0153 60C3 17 022E
GR82 LBSR GETDAT      0154 60C6      B6      6007      LDA
RESDAT      0155 60C9 43      COMA      0156
60CA 17 028E      LBSR DSPBIN disp response      0157 60CD 5A
DECB      0158 60CE 26      F3 BNE GR82      0159
0160 60D0      BD      6384      JSR CRLF      0161 *
0162 *----- cmd 58 -----      0163 60D3      SCMD58
0164 60D3 17 0273      LBSR SETCKL      0165 60D6      C6
06      LDB      #6      0166 60D8 8E 6427      LDX
#CMD58      0167 60DB      A6      80      SC581 LDA      , X +
0168 60DD 43      COMA      0169 60DE 17 01F7
LBSR SNDDAT      0170 60E1 5A      DECB
0171 60E2 26      F7 BNE SC581      0172 * get response
0173 60E4      GR580      0174 60E4 17 020D      LBSR
GETDAT      0175 60E7      B6      6007      LDA RESDAT
0176 60EA 43      COMA      0177 60EB 17 026D
LBSR DSPBIN disp response      0178 60EE 81 01      CMPA      #$01
0179 60F0 26      F2 BNE GR580      0180 60F2      C6      04      GR581 LDB
#4      0181 60F4 17 01FD      GR582 LBSR GETDAT      0182
60F7      B6      6007      LDA RESDAT      0183 60FA 43
COMA      0184 60FB 17 025D      LBSR DSPBIN disp response
0185 60FE 5A      DECB      0186 60FF 26      F3 BNE
GR582      0187      0188 6101      RPT41
0189 *----- cmd 55 -----      0190 6101      SCMD55
0191 6101 17 0245      LBSR SETCKL      0192 6104      C6
06      LDB      #6      0193 6106 8E 642D      LDX
#CMD55      0194 6109      A6      80      SC551 LDA      , X +
0195 610B 43      COMA      0196 610C 17 01C9
LBSR SNDDAT      0197 610F 5A      DECB
0198 6110 26      F7 BNE SC551      0199 * get response
0200 * CLRB      0201 6112      GR550      0202 * DECB
0203 * BEQ SDERR      0204 6112 17 01DF      LBSR GETDAT
0205 6115      B6      6007      LDA RESDAT      0206 6118 43

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COMA                                0207 6119 17 023F                                LBSR DSPBIN disp response
0208 611C 81 01                    CMPA      #$01                                0209 611E 26      F2
BNE GR550                          0210                                0211 6120      BD      6384
JSR CRLF                          0212 *----- acmd 41 -----                                0213 6123
ASCMD41                            0214 6123 17 0223                                LBSR SETCKL
0215 6126      C6      06                                LDB      #6                                0216 6128 8E 6433
LDX      #ACMD41                                0217 612B      A6      80      ASC411 LDA      , X +
0218 612D 43                                COMA                                0219 612E 17 01A7
LBSR SNDDAT                        0220 6131 5A                                DECB
0221 6132 26      F7 BNE ASC411                                0222 * get response
0223 6134                                GR410                                0224 6134 17 01BD                                LBSR
GETDAT                            0225 6137      B6      6007                                LDA RESDAT
0226 613A 43                                COMA                                0227 613B 17 021D
LBSR DSPBIN disp response          0228 613E 17 01B3                                LBSR GETDAT
0229 6141      B6      6007                                LDA RESDAT                                0230 6144 43
COMA                                0231 6145 17 0213                                LBSR DSPBIN disp response
0232 6148 81 00                    CMPA      #$00                                0233 614A 26      B5
BNE RPT41                          0234                                0235 614C      BD      6384
JSR CRLF                          0236                                0237 *----- second cmd 58 -----
0238 614F      SCMD258                                0239 614F 17 01F7                                LBSR
SETCKL                            0240 6152      C6      06                                LDB      #6
0241 6154 8E 6427                    LDX      #CMD58                                0242 6157      A6
80      SC2581 LDA      , X +                                0243 6159 43                                COMA
0244 615A 17 017B                    LBSR SNDDAT                                0245 615D 5A
DECB                                0246 615E 26      F7 BNE SC2581                                0247 * get
response                          0248 6160                                GR2580                                0249 6160 17
0191                                LBSR GETDAT                                0250 6163      B6      6007
LDA RESDAT                        0251 6166 43                                COMA
0252 6167 17 01F1                    LBSR DSPBIN disp response          0253 616A 81 00
CMPA      #$00                                0254 616C 26      F2 BNE GR2580
0255 616E 17 0183                    LBSR GETDAT                                0256 6171      B6
6007                                LDA RESDAT                                0257 6174 43                                COMA
0258 6175 17 01E3                    LBSR DSPBIN disp response          0259 6178 48
LSLA                                0260 6179 48                                LSLA
0261 617A 24 03                    BCC GR2581                                0262 617C 7C 6008
INC SDHC                          if sdhc then 1                                0263 617F      C6      03      GR2581 LDB
#3                                0264 6181 17 0170      GR2582 LBSR GETDAT                                0265
6184      B6      6007                                LDA RESDAT                                0266 6187 43
COMA                                0267 6188 17 01D0                                LBSR DSPBIN disp response
0268 618B 5A                                DECB                                0269 618C 26      F3 BNE
GR2582                            0270                                0271 618E 8E 648E                                LDX
#MSSD0                            0272 6191      B6      6008                                LDA SDHC
0273 6194 27 03                    BEQ SD1                                0274 6196 8E 6496
LDX      #MSSD1                                0275 6199      BD FAA4 SD1 JSR PSTRNG
0276                                0277 619C 39                                RTS                                0278
0279 *****                                0280 * GET SD DATA
ROUTINE                          0281 * use CMD9                                0282
*****                                0283 619D                                GETSD
EQU      *                                0284 619D 17 01A9                                LBSR SETCKL

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0285 61A0      C6      06          LDB      #6          0286 61A2 8E 6439
LDX      #CMD9          0287 61A5      A6      80      SC91 LDA      , X +
0288 61A7 43          COMA          0289 61A8 17 012D
LBSR SNDDAT          0290 61AB 5A          DECB
0291 61AC 26      F7 BNE SC91          0292 61AE      BD      6384
JSR CRLF          0293 * get response          0294 61B1          GR90
0295 61B1 17 0140          LBSR GETDAT          0296 61B4      B6
6007          LDA RESDAT          0297 61B7 43          COMA
0298 61B8 17 01A0          LBSR DSPBIN disp response          0299 61BB 81 00
CMPA      #$00          0300 61BD 26      F2 BNE GR90
0301          0302 61BF      BD      6384          JSR CRLF
0303 61C2      C6      14          LDB      #16+1+2+1 fe,crcH,crcL,ff 0304 61C4
17 012D      GR91 LBSR GETDAT          0305 61C7      B6      6007
LDA RESDAT          0306 61CA 43          COMA
0307 61CB 17 018D          LBSR DSPBIN disp response          0308 61CE 5A
DECB          0309 61CF 26      F3 BNE GR91          0310 61D1
39          RTS          0311          0312 *
0313 61D2          SDERR EQU      *          0314 61D2 8E 64A0
LDX      #MERR          0315 61D5      BD FAA4 JSR PSTRNG
0316 61D8          OKEND          0317 61D8 7E      F800 JMP MON
0318 *          0319 *****
0320 * READ ROUTINE          0321 * use CMD17          0322 * set sctno
to $6005 , 6006          0323 * read data set to $6800 - 69FF
0324 *****          0325 61DB
SDRD EQU      *          0326 61DB      BD      6384          JSR CRLF
0327 61DE 17 012D          LBSR INSCT          0328 * copy sct no to
cmd param          0329 61E1 8E 6003          LDX      #SCTN0
0330 61E4 108E 6416          LDY      #CMD17+1          0331 61E8
C6      04          LDB      #4          0332 61EA      A6      80      CPY1
LDA      , X +          0333 61EC      A7 A0 STA      , Y +          0334
61EE 5A          DECB          0335 61EF 26      F9 BNE CPY1
0336 * send cmd          0337 61F1 17 0155          LBSR SETCKL
0338 61F4      C6      06          LDB      #6          0339 61F6 8E 6415
LDX      #CMD17          0340 61F9      A6      80      SDR1 LDA      , X +
0341 61FB 17 015D          LBSR DSPBIN disp response          0342 61FE 43
COMA          0343 61FF 17 00D6          LBSR SNDDAT
0344 6202 5A          DECB          0345 6203 26      F4 BNE
SDR1          0346 * get response          0347 6205          SDR2
0348 6205 17 00EC          LBSR GETDAT          0349 6208      B6
6007          LDA RESDAT          0350 620B 43          COMA
0351 620C 17 014C          LBSR DSPBIN disp response          0352 620F 81
FE CMPA      #$FE token?          0353 6211 26      F2 BNE SDR2          0354
0355 * get data          0356 6213      BD      6384          JSR CRLF
0357 6216 8E 6600          LDX      #RWBUF <-temp buf          0358 6219 108E
0200          LDY      #512          0359 621D      C6      20
DSPDAT LDB      #32          0360 621F 31 3F      DSPD1 LEAY      - 1 , Y
0361 6221 17 00D0          LBSR GETDAT          0362 6224      B6
6007          LDA RESDAT          0363 6227 43          COMA
0364 6228      A7      80          STA      , X +          save buf 0365

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```

622A 17 012E          LBSR DSPBIN disp data          0366 622D 108C 0000
CMPLY      #0          0367 6231 27 08          BEQ DSPD2
0368 6233 5A          DECB          0369 6234 26      E9 BNE
DSPD1          0370 6236      BD      6384          JSR CRLF
0371 6239 20      E2 BRA DSPDAT          0372          0373 623B      BD
6384      DSPD2 JSR CRLF          0374 623E 17 00B3          LBSR GETDAT
0375 6241      B6      6007          LDA RESDAT          0376 6244 43
COMA          0377 6245 17 0113          LBSR DSPBIN crch
0378 6248 17 00A9          LBSR GETDAT          0379 624B      B6
6007          LDA RESDAT          0380 624E 43          COMA
0381 624F 17 0109          LBSR DSPBIN crcl          0382 6252 39
RTS          0383          0384
*****
0385 * WRITE ROUTINE
0386 * use CMD24          0387 * set sctno to $6005 , 6006          0388
* write data from $6A00 - 6BFF          0389
*****
0390 6253          SDWRT
EQU      *          0391 6253 8E 6482          LDX      #MSWROK
0392 6256      BD FAA4 JSR PSTRNG          0393 6259      BD F000 JSR INCHA
0394 625C 84      DF ANDA      #$DF          0395 625E 81 59
CMPA      #'Y          0396 6260 27 01          BEQ SDWR1
0397 6262 39          RTS          0398 6263      BD      6384
SDWR1 JSR CRLF          0399 6266 17 00A5          LBSR INSCT
0400 * copy sct no to cmd param          0401 6269 8E 6003          LDX
#SCTN0          0402 626C 108E 641C          LDY      #CMD24+1
0403 6270      C6      04          LDB      #4          0404 6272      A6
80      CPY2 LDA      , X +          0405 6274      A7 A0 STA      , Y +
0406 6276 5A          DECB          0407 6277 26      F9 BNE
CPY2          0408 * send cmd          0409 6279 17 00CD
LBSR SETCKL          0410 627C      C6      06          LDB      #6
0411 627E 8E 641B          LDX      #CMD24          0412 6281      A6
80      SDW1 LDA      , X +          0413 6283 17 00D5          LBSR DSPBIN
disp response          0414 6286 43          COMA
0415 6287 8D 4F          BSR SNDDAT          0416 6289 5A
DECB          0417 628A 26      F5 BNE SDW1          0418 * get
cmd response          0419 628C          SDW2          0420 628C
8D 66          BSR GETDAT          0421 628E      B6      6007
LDA RESDAT          0422 6291 43          COMA
0423 6292 17 00C6          LBSR DSPBIN disp response          0424 6295 81 00
CMPA      #$00          0425 6297 26      F3 BNE SDW2
0426 * 1 byte skip          0427 6299 86      FF LDA      #$FF
0428 629B 43          COMA          0429 629C 8D 3A
BSR SNDDAT          1byte skip          0430 629E      BD      6384          JSR
CRLF          0431          0432 * send data packet          0433
62A1 8E 6600          LDX      #RWBUF <-temp buf          0434 62A4 108E 0200
LDY      #512          0435 * send data token          0436 62A8
86      FE LDA      #$FE          0437 62AA 43          COMA
0438 62AB 8D 2B          BSR SNDDAT data token          0439 62AD      A6      80
SNDDT1 LDA      , X +          0440 62AF 43          COMA
0441 62B0 8D 26          BSR SNDDAT          0442 62B2 31 3F

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```

LEAY      - 1 , Y                                0443 62B4 26      F7 BNE SNDDT1
0444                                0445 62B6 86      FF LDA      #$FF                                0446 62B8 43
COMA                                0447 62B9 8D 1D                                BSR SNDDAT crch
0448 62BB 8D 1B                                BSR SNDDAT crcl                                0449 * get data response
0450 62BD                                CKRS                                0451 62BD 8D 35                                BSR
GETDAT                                0452 62BF      B6      6007                                LDA RESDAT
0453 62C2 43                                COMA                                0454 62C3 17 0095
LBSR DSPBIN disp response                                0455 62C6 81      E5 CMPA      #$E5 data response?
0456 62C8 26      F3 BNE CKRS                                0457 * wait until traffic is over
0458 62CA                                BUSY                                0459 62CA 8D 28                                BSR
GETDAT                                0460 62CC      B6      6007                                LDA RESDAT
0461 62CF 43                                COMA                                0462 62D0 17 0088
LBSR DSPBIN disp response                                0463 62D3 81      FF CMPA      #$FF
0464 62D5 26      F3 BNE BUSY                                0465 62D7 39                                RTS
0466 *****                                0467
*===== sub routine =====                                0468 * send data
0469 62D8                                SNDDAT EQU      *                                0470 62D8 34 06
PSHS A , B                                0471 62DA 7F 6007                                CLR RESDAT
0472 62DD      C6      08                                LDB      #8                                0473 62DF 34 02
SNDD1 PSHS A                                0474 62E1 84      FC ANDA      #$FC
0475 62E3 8A 03                                ORA      #$03 miso:b1=H(for PIA),cs:b0=L                                0476
62E5      B7 F0DC STA PADAT send data                                0477 62E8 8D 5F                                BSR SETCKL
0478 62EA 35 02                                PULS A                                0479 62EC 48
LSLA                                0480 62ED 8D 63                                BSR SETCKH
0481 62EF 5A                                DECB                                0482 62F0 26      ED BNE
SNDD1                                0483 62F2 35 86                                PULS A , B , PC
0484                                0485 * get data                                0486 62F4                                GETDAT EQU
*                                0487 62F4 34 06                                PSHS A , B
0488 62F6      C6      08                                LDB      #8                                0489 62F8 8D 4F
GETD1 BSR SETCKL                                0490 62FA 12                                NOP
0491 62FB 12                                NOP                                0492 62FC 12
NOP                                0493 62FD 8D 53                                BSR SETCKH
0494 62FF      B6 F0DC LDA PADAT                                get response                                0495 6302 44
LSRA                                0496 6303 44                                LSRA b1 -> C                                0497
6304 79 6007                                ROL RESDAT C -> b0                                0498 6307 5A
DECB                                0499 6308 26      EE BNE GETD1                                0500 630A
8D 3D                                BSR SETCKL                                0501 630C 35 86                                PULS
A , B , PC                                0502 *                                0503                                0504 * input
SCT No                                0505 630E                                INSCT EQU      *                                0506
630E 8E 647A                                LDX      #MSSCT                                0507 6311      BD FAA4
JSR PSTRNG                                0508 6314 8D 79                                BSR GET8CH                                get 8chr ,
convert 4bin                                0509                                0510 6316 7D 6008                                TST SDHC
0511 6319 26 1D                                BNE CALCHC                                0512 * SD card
0513 631B      C6      03                                LDB      #3                                0514 631D 8E 6004
LDX      #SCTN1                                0515 6320 108E 6003                                LDY      #SCTN0
0516 6324      A6      80      SFT1 LDA      , X +                                0517 6326      A7 A0
STA      , Y +                                0518 6328 5A                                DECB
0519 6329 26      F9 BNE SFT1                                0520 632B 7F 6006                                CLR
SCTN3                                0521 * calc top adrs of sct                                0522 632E

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CALCSD EQU      *                0523 632E 78 6006                LSL SCTN3
0524 6331 79 6005                ROL SCTN2                0525 6334 79 6004
ROL SCTN1                0526 6337 39                RTS
0527                0528 * SDHC card                0529 * calc top adrs of sct
0530 6338                CALCHC EQU      *                0531 6338 8E 6003
LDX      #SCTN0                0532 633B 108E 6416                LDY
#CMD17+1                0533 633F      C6      04                LDB      #4
0534 6341      A6      80      CALCH1 LDA      , X +                0535 6343      A7 A0
STA      , Y +                0536 6345 5A                DECB
0537 6346 26      F9 BNE CALCH1                0538 6348 39                RTS
0539                0540 *                0541 * set sck : L                0542 6349
SETCKL EQU      *                0543 6349 34 02                PSHS A
0544 634B 86 3C                LDA      #$3C ca2=L                0545 634D      B7 F0DD STA
PACTL                0546 6350 35 82                PULS A , PC
0547                0548 * set sck : H                0549 6352                SETCKH EQU
*                0550 6352 34 02                PSHS A
0551 6354 86 34                LDA      #$34 ca2=H                0552 6356      B7 F0DD STA
PACTL                0553 6359 35 82                PULS A , PC
0554                0555 * binary ( A )-> ascii ( A )( B )                0556 635B
DSPBIN EQU      *                0557 635B 34 06                PSHS A , B
0558 635D 1F 89                TFR A , B                0559 635F 44
LSRA                0560 6360 44                LSRA
0561 6361 44                LSRA                0562 6362 44
LSRA                0563 6363 8A 30                ORA      #$30
0564 6365      C4      0F                ANDB      #$0F                0565 6367      CA
30                ORB      #$30                0566 6369 81 39                CMPA
#$39                0567 636B 23 02                BLS BI21
0568 636D 8B 07                ADDA      #$07                0569 636F      C1
39      BI21 CMPB      #$39                0570 6371 23 02                BLS BI22
0571 6373      CB      07                ADDB      #$07                0572 6375      BD
F002 BI22 JSR OUTCHA                0573 6378 1F 98                TFR B , A
0574 637A      BD F002 JSR OUTCHA                0575 637D 86 20                LDA
#$20                0576 637F      BD F002 JSR OUTCHA                0577 6382 35
86                PULS A , B , PC                0578                0579 * crlf
0580 6384                CRLF EQU      *                0581 6384 86 0D
LDA      #$0D                0582 6386      BD F002 JSR OUTCHA
0583 6389 86 0A                LDA      #$0A                0584 638B      BD
F002 JSR OUTCHA                0585 638E 39                RTS
0586                0587 *                0588 638F                GET8CH EQU      *
0589 * input chr                0590 638F 8E 64C6                LDX      #INPBUF
0591 6392 5F                CLRB                0592 6393      BD F000 CLOP
JSR INCHA                0593 6396 81 0D                CMPA      #$0D
0594 6398 27 07                BEQ INEND                0595 639A      A7      85
STA B , X                0596 639C 5C                INCB
0597 639D      C1      08                CMPB      #8                0598 639F 25      F2
BLO CLOP                0599                0600 63A1                INEND
0601 * shift                0602 63A1 5D                TSTB
0603 63A2 27 11                BEQ CL1                0604 63A4 8E 64C6
LDX      #INPBUF                0605 63A7 3A                ABX

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0606 63A8 108E 64CE          LDY          #INPBUF+8          0607 63AC 34
04          PSHS B          0608 63AE          A6          82          SFL0P
LDA          , -X          0609 63B0          A7 A2 STA          , - Y          0610
63B2 5A          DECB          0611 63B3 26          F9 BNE SFL0P
0612 * clr          0613 63B5          C6          08          CL1 LDB          #8
0614 63B7          E0 E0 SUBB          , S +          0615 63B9 27 0A          BEQ
CLEND          0616 63BB 8E 64C6          LDX          #INPBUF
0617 63BE 86 30          LDA          #'0          0618 63C0          A7
80          CLL0P STA          , X +          0619 63C2 5A          DECB
0620 63C3 26          FB BNE CLL0P          0621 63C5          CLEND
0622          0623 * convert to bin          0624 63C5          C6          04
LDB          #4          0625 63C7 8E 64C6          LDX          #INPBUF
0626 63CA 108E 6003          LDY          #SCTN0          0627 63CE
CVT1          0628 63CE          A6          80          LDA          , X +
0629 63D0 8D 1F          BSR CVBIN          0630 63D2 25 16
BCS EREND1          0631 63D4 48          LSLA
0632 63D5 48          LSLA          0633 63D6 48          LSLA
LSLA          0634 63D7 48          LSLA
0635 63D8 34 02          PSHS A          0636 63DA          A6          80
LDA          , X +          0637 63DC 8D 13          BSR CVBIN
0638 63DE 25 08          BCS EREND0          0639 63E0          AB E0 ADDA
, S +          0640 63E2          A7 A0 STA          , Y +          0641 63E4 5A
DECB          0642 63E5 26          E7 BNE CVT1          0643 63E7
39          RTS          0644          0645 63E8 35 02
EREND0 PULS A          0646 63EA 8E 64B7          EREND1 LDX          #MSERR
0647 63ED          BD FAA4 JSR PSTRNG          0648 63F0 39          EEND RTS
0649          0650 *-----          0651 * ascii -> bin
0652 63F1          CVBIN EQU          *          0653 63F1 80 30
SUBA          #'0          0654 63F3 2B 11          BMI CVERR
0655 63F5 81 09          CMPA          #9          0656 63F7 23 0A
BLS CVBEND          0657 63F9 81 11          CMPA          #$11
0658 63FB 25 09          BLO CVERR          0659 63FD 81 16
CMPA          #$16          0660 63FF 22 05          BHI CVERR
0661 6401 80 07          SUBA          #7          0662 6403
CVBEND          0663 6403 1C          FE ANDCC          #$FE reset C          0664 6405
39          RTS          0665 6406 1A 01          CVERR ORCC
#$01 set C          0666 6408 39          RTS          0667
*****          0668 *
0669 *===== parameter =====          0670 6409 40          CMD0 FCB $40
, $00 , $00 , $00 , $00 , $95          '@          0671 640F 41 CMD1 FCB
$41,$00,$00,$00,$00,$FF ' A          0672 6415 51          CMD17 FCB $51 , $00 , $00 ,
$00 , $00 , $FF          'Q          0673 641B 58 CMD24 FCB $58,$00,$00,$00,$00,$FF '
X          0674 6421 48          CMD8 FCB $48 , $00 , $00 , $01 , $AA , $87          'H
0675 6427 7A CMD58 FCB $7A,$00,$00,$00,$00,$FF ' z          0676 642D 77          CMD55
FCB $77 , $00 , $00 , $00 , $00 , $FF          'w          0677 6433 69 ACMD41 FCB
$69,$40,$FF,$80,$00,$FF ' i          0678 6439 49          CMD9 FCB $49 , $00 , $00 , $00
, $00 , $FF          'I          0679          0680 *0681 643F 31 MSMENU FCC
/1:Read Sct, 2:Write Sct, 3:Get SD data, 4:SD Init, 9:End: /          0682 6479 04 FCB 4
0683 647A 53 MSSCT FCC /SctNo: /          0684 6481 04 FCB 4          0685 6482 57

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MSWROK FCC /Write OK?: /          0686 648D 04 FCB 4          0687 648E 53 MSSD0
FCC /SD Card/          0688 6495 04 FCB 4          0689 6496 53 MSSD1 FCC /SDHC
Card/          0690 649F 04 FCB 4          0691 64A0 53 MERR FCC /SD Initialize
error!/, $0D, $0A      0692 64B6 04 FCB 4          0693 64B7 49 MSERR FCC /Input
error!/, $0D, $0A      0694 64C5 04 FCB 4          0695
*****0696 64C6 FF INPBUF RMB 8          0697
0698 END START          0 error(s), 0 warning(s) is detected. Label addressACMD41
6433 ASC411 612B ASCMD41 6123 BI21 636F BI22 6375BUSY 62CA CALCH1 6341 CALCHC 6338
CALCSD 632E CKRS 62BDCL1 63B5 CLEND 63C5 CLL0P 63C0 CLOP 6393 CMD0 6409CMD1 640F CMD17
6415 CMD24 641B CMD55 642D CMD58 6427CMD8 6421 CMD9 6439 CPY1 61EA CPY2 6272 CRLF
6384CVBEND 6403 CVBIN 63F1 CVERR 6406 CVT1 63CE DSPBIN 635BDSPD1 621F DSPD2 623B
DSPDAT 621D EEND 63F0 EREND0 63E8EREND1 63EA GET8CH 638F GETD1 62F8 GETDAT 62F4 GETSD
619DGR00 608C GR2580 6160 GR2581 617F GR2582 6181 GR410 6134GR550 6112 GR580 60E4
GR581 60F2 GR582 60F4 GR80 60B3GR81 60C1 GR82 60C3 GR90 61B1 GR91 61C4 INCHA F000INEND
63A1 INIT 602D INPBUF 64C6 INSCT 630E ISTT 6071MAIN 6009 MEN2 6044 MEN3 604D MEN4 6056
MEN5 605FMENU 602F MERR 64A0 MON F000 MSERR 64B7 MSMENU 643FMSSCT 647A MSSD0 648E
MSSD1 6496 MSWROK 6482 OKEND 61D8OUTCHA F002 PACTL F0DD PADAT F0DC PSTRNG FAA4 RESDAT
6007RPT41 6101 RWBUF 6600 SC01 6082 SC2581 6157 SC551 6109SC581 60DB SC80 60AA SC91
61A5 SCMD0 607A SCMD258 614FSCMD55 6101 SCMD58 60D3 SCMD8 60A2 SCTN0 6003 SCTN1
6004SCTN2 6005 SCTN3 6006 SD1 6199 SDERR 61D2 SDHC 6008SDINIT 6069 SDR1 61F9 SDR2 6205
SDRD 61DB SDW1 6281SDW2 628C SDWR1 6263 SDWRT 6253 SETCKH 6352 SETCKL 6349SFLOP 63AE
SFT1 6324 SNDD1 62DF SNDDAT 62D8 SNDDT1 62ADSTART 6000 VN 6002

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This is the program's menu section.

1:Read Sct, 2:Write Sct, 3:Get SD data, 4:SD Init, 9:End:

When you select menu option 1 or 2, it will ask for a sector number. Entering this number will either read the contents of that sector or write the contents of the buffer to that sector.

You can also display SD card information in step 3.

The INCHA, OUTCHA, and PSTRNG subroutines are used during monitoring, but there's nothing else special about them, so with a 6809 machine and a 6821, you should be able to read and write to an SD card.